Md **Raihan** Ahmed

Software Engineer · Security Researcher

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"Let me help you make the change that you want to see in this world."

Experience

Meta — *Menlo Park, California, USA*

Software Engineer Intern — *May 2025 - Aug. 2025*

- Pinpointed vulnerabilities in manual asset classification, automated risk detection via ML model to improve data privacy by **11.25%.**
- Engineered 5 ML models to strengthen Meta security compliance across cloud infrastructure aligned to NIST SP800-53.
- Revolutionized data security measures by selecting the best-performing risk-based prioritization model, achieving an **89% F1 score.**
- Deployed high-fidelity ML model via continuous integration (CI/CD) pipelines, improving **precision and recall by 89%.**
- Accelerated the security data aggregation by 99.72%, significantly reducing the timeline from **months to a fraction of the time**

University of Utah — *Utah, USA*

Research Assistant — *Aug. 2021 - Current*

- Led threat intelligence research on Industrial Control systems, found exposure to **advanced persistent threats** using MITRE TTPs.
- Architected ICSTracker (LLVM/C++) to capture and analyze large-scale unstructured data sets to identify attacker footprints.
- Transformed unstructured log data into provenance graph and reduced size by **97.71% (82K to 43)** via 3 algorithmic steps.
- Simulated 10 **MITRE ATT&CK** cyberattacks cross-compiled for ARM-32/ARM-64 emulation to validate resilience.
- Achieved **56%** higher attack detection coverage compared to all prior methods across 10 attack scenarios on 2 testbeds.

University of Utah — *Utah, USA*

Teaching Assistant

- TA for CS 4400 (Computer Systems) and CS 6956 (Software & Systems Security), mentored 130+ students in Fall'22,24, Spring'23
- Taught OS internals, security concepts, e.g., buffer overflow attacks, and improved their thought process on systems thinking.

Projects

AI Agent: Network Vulnerability Scanning (GenAI Security) — *Python, LLMs*

Personal Project — *GitHub* — *Oct. 2025 – Nov. 2025*

- Spearheaded 2 GenAI agents for automated network security scanning, enabling authorized discovery of vulnerabilities.
- Detection of 1 critical Remote Code Execution could improve system efficiency by 10–30%, <5% for minor misconfiguration.

Intrusion Detection(IDS/IPS) — *C++, LTL, LLVM*

Personal Project — *Paper* — *Aug 2021 – May 2024*

- Implemented an LTL-based provenance analysis system detecting 4 unique cyberattacks in Industrial Control Systems.
- Demonstrated 100% detection and full coverage, validating system robustness to facilitate Security Incident Response.

Fuzzing XPDF-4.05 — *LLVM, C++*

Applied S/W Security Test. Course — *Git1* *Git2* — *Aug. 2022 - Dec. 2022*

- Extended fuzzing campaign with static analysis (control flow + IR inspection) to guide fuzzing input generation.
- Applied program verification tools (Z3, CBMC) to validate discovered crash paths.
- Improved open-source robustness by contributing bug reports and patches.

Rollback attack in Cars — *Arduino, C++*

Cyber-phys Sys & IoT Security Course — *Github* — *Aug. 2022 - Dec. 2022*

- Formulated project was to build a device to perform cyberattack on modern cars.
- Built a rollback device with Arduino to access cars without using keys.

ICSTracker: To Trace back to the source of Intrusion in Industrial IoT Systems. — *C++, LLVM, Python*

PhD 1st Proj — *GitHub* — *Aug. 2021 - May 2024*

- Created an instrumentation tool with LLVM to trace back to the source of attack.
- Physical process-aware, cross-iteration and cross-domain approach to backtracking intrusions in ICS
- Crafted detailed instrumentation passes to capture critical runtime data from ICS controllers; improved data collection accuracy by 56%, to enable more effective troubleshooting for complex system failures.

Fake News Prediction (ML Detection) — *Python, Scikit-Learn*

PERSONAL PROJECT — *GitHub* — *Mar. 2025 – Apr. 2025*

- Architected and tuned 4 ML models to detect anomalous patterns in large text datasets, securing 99.9% prediction accuracy.
- Utilized Gradient Boosting to effectively capture complex data patterns in unstructured data.

Binary Image Classification — *Python, TensorFlow*

PERSONAL PROJECT — *GitHub* — *Mar. 2025 – Apr. 2025*

- Trained 1 convolutional neural network to classify images improving model accuracy via hyper-parameter tuning and rescaling.
- Attained 80% accuracy with reduced overfitting using image rescaling, binary cross-entropy loss.

Remote control robot built with Arduino — *Arduino, C++*

A PERSONAL COOL PROJECT — *Demo* — *May 2021*

- Used arduino uno to build this robot and controlling it with Bluetooth

Interactive Visualization tool — *Javascript*

DATA VISUALIZATION COURSE — *GitHub* — *Aug. 2022 - Dec. 2022*

- Formulated project is to build an interactive tool for video games.
- Built website with javascript to explore video game sales by year and platform.

Publications

IEEE/IFIP International Conference on Dependable Systems and Networks — *Naples, Italy*

ICSTRACKER: BACKTRACKING INTRUSIONS IN MODERN INDUSTRIAL CONTROL SYSTEMS **(1ST AUTHOR)** — *paper* — *Apr. 2025*

- Recovered program semantics from highly complex Industrial Control Systems to relate source & destination of a cyberattack.
- Reconstructed data dependencies between the system to application level events to define the provenance model.
- Linked controller operations to OS- level events to show the attacker footprints of a cyber-physical attacks in ICS/CPS setting.

Re-design Industrial Control Systems with Security (RICSS) — *Utah, USA*

CONTEXT-AWARE INTRUSION DETECTION IN INDUSTRIAL CONTROL SYSTEMS **(1ST AUTHOR)** — *paper* — *Oct. 2024*

- Industrial Control Systems (ICS) suffer from high false positives and poor detection of Advanced Persistent Threats (APTs).
- Designed a detection framework capable of capturing fine-grained, context-aware attack behaviors in ICS environments.
- Developed a provenance-based intrusion detection system with Linear Temporal Logic(LTL) to enhance ICS attack detection.
- Enabled improved differentiation between benign and malicious activities, significantly reducing false positives.

Electrical, Computer and Communication Engineering (ECCE) — *Bangladesh*

A COMPARATIVE ANALYSIS OF TRADITIONAL AND MODERN DATA COMPRESSION SCHEMES FOR LARGE MULTIDIMENSIONAL EXTENDIBLE ARRAY **(5TH AUTHOR)** — *paper* — *Feb. 2019*

- Traditional storage methods become inefficient as dataset dimensionality increases (1D → 3D MOLAP systems).
- Evaluated how compression techniques scale across different dimensions.
- Analyzed compression schemes across multi-dimensional array structures, measuring compression ratio and space efficiency.
- Identified scalability and validated that advanced schemes (EaCRS) outperform traditional methods in higher dimensions.

Biointerface Research in Applied Chemistry — *Bangladesh*

OVARIAN CANCER SUBSTANTIAL RISK FACTOR ANALYSIS BY MACHINE LEARNING: A LOW INCOMING COUNTRY **(1ST)** — *paper* — *Jul. 2020*

- Ovarian cancer remains a leading cause of death among women, with limited early-stage risk identification.
- Identify and rank significant risk factors using real-world clinical data collected from Hospitals of Dhaka, Bangladesh.
- Collected and preprocessed a dataset of 521 patients (47 features) and applied statistical + machine learning analysis.
- Identified 25 high-impact risk factors (e.g., abortions, menopause issues), enabling improved early-risk assessment.

BMC bioinformatics — *Bangladesh*

MACHINE LEARNING TO REVEAL AN ASTUTE RISK PREDICTIVE FRAMEWORK FOR GYNECOLOGIC CANCER AND ITS IMPACT ON WOMEN PSYCHOLOGY: BANGLADESHI PERSPECTIVE **(B.SC PROJ, 2ND AUTHOR)** — *GitHub* — *paper* — *Apr. 2021*

- Found high incidence of cervical and ovarian cancer among women, with limited awareness and high stress levels.
- Develop a predictive model to identify risk factors and estimate cancer risk, considering the psychological impact of stress.
- Collected and preprocessed data from 866 patients, and trained multiple machine learning models e.g., SVM, AdaBoost etc.
- Developed an application with ML model to predict risk levels of cervical and ovarian cancer trained based on the risk factors.

Elsevier — *Bangladesh*

A BIOINFORMATICS APPROACH FOR IDENTIFICATION OF THE CORE ONTOLOGIES AND SIGNATURE GENES OF PULMONARY DISEASE AND ASSOCIATED DISEASES **(2ND AUTHOR)** — *paper* — *Jul. 2020*

- Gynecologic cancers remain difficult to predict due to complex genetic, environmental, and clinical factors.
- Develop a ML model to identify risk factors and predict susceptibility to gynecologic cancers with higher accuracy.
- Collected multi-dimensional patient data, preprocessed it, and applied machine learning algorithms.
- Produced a risk prediction framework that highlights significant predictors and provides a robust tool for early identification.

Journal of Proteins and Proteomics *Bangladesh*

PROTEIN INTERACTION NETWORK AND DRUG DESIGN OF STOMACH CANCER AND ASSOCIATED DISEASE: A BIOINFORMATICS APPROACH **(1ST AUTHOR)** *paper* *Dec. 2020*

- Identified common genes among seven diseases (OB, SC, CC, PC, PRC, LK, MD) to analyze their biological and genetic network.

Education

University of Utah *Utah, USA*

PH.D. IN COMPUTER SCIENCE, **GPA: 3.86** *Aug. 2021 - Aug. 2026 (Expected)*

University of Utah *Utah, USA*

MS. IN COMPUTER SCIENCE, **GPA: 3.86** *Aug. 2021 - May 2024*

Skills

Security Domains	Threat detection & monitoring, incident response, vulnerability management, IDS/IPS, SIEM
Security Tools	AFL++, Z3, CBMC, Snort, Suricata, OSS-Fuzz, Provenance Tracking
Cyber Defense	Threat Intelligence, Threat Modeling, endpoint security, IAM, AI/LLM security
Dev. & Deployment Tools	LLVM Compiler Tools, GitHub, Docker
Programming Lang.	Python, C/C++, Javascript, C#, JAVA, PHP
Frameworks	MITRE ATT&CK, Cyber Kill Chain, React.js, Node.js, Express.js, Android Studio, Dot(.) Net
Data Science & ML	Tensorflow, NumPy, scikit-learn, Pandas, Matplotlib
Database	Oracle SQL, MySQL
Data Analysis Software	Matlab, Google Colab, Jupyter, Anaconda
Cloud & Infrastructure	AWS (EC2, S3, IAM), Docker, CI/CD pipelines, cloud-native security, Kubernetes
Interests	Security, Privacy, Machine learning, Artificial Intelligence